



National Comprehensive
Cancer Network®

2026

NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the
National Comprehensive Cancer Network® (NCCN®)

Acute Lymphoblastic Leukemia



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NCCN Guidelines for Patients®

The essential guide for people facing cancer.

Based on care recommendations from leading cancer experts.

Explains high-quality cancer care provided at
state-of-the-art cancer centers.

Reviewed and revised every year.

Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Because cancer care is always evolving, NCCN develops and frequently updates evidence-based cancer care recommendations used by health care providers worldwide. These recommendations are known as the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The NCCN Guidelines for Patients plainly explain these expert recommendations, so you can talk with your care team about the best care for you.

**These NCCN Guidelines for Patients are based on the NCCN Guidelines®
for Acute Lymphoblastic Leukemia, Version 1.2026 — April 8, 2026.**

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Contents

4	Chapter 1 About ALL	46	Chapter 7 T-ALL treatments
9	Chapter 2 Testing to diagnose ALL	52	Chapter 8 Supportive care
14	Chapter 3 Testing to plan treatment for ALL	59	Chapter 9 Other resources
23	Chapter 4 Phases of treatment	63	Words to know
28	Chapter 5 Types of treatment	65	NCCN Contributors
38	Chapter 6 Ph-negative and Ph-positive B-ALL treatments	66	NCCN Cancer Centers
		68	Index

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NCCN Foundation seeks to support the millions of patients and their families affected by a cancer diagnosis by funding and distributing NCCN Guidelines for Patients. NCCN Foundation is also committed to advancing cancer treatment by funding the nation's promising doctors at the center of innovation in cancer research. For more details and the full library of patient and caregiver resources, visit [NCCN.org/patients](https://www.nccn.org/patients).

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1

About ALL

- 5 What is ALL?
- 6 What are the types of ALL?
- 7 How is ALL treated?
- 8 What is supportive care?
- 8 How can I get the best care?

Acute lymphoblastic leukemia (ALL) is a fast-growing cancer that starts in lymphocytes (a type of white blood cell). Treatment depends on the type of ALL you have, your age at diagnosis, and other factors. This guide is for people with ALL who are being treated at an adult cancer center.

What is ALL?

Acute lymphoblastic leukemia (ALL) is a blood cancer that starts in disease-fighting white blood cells called lymphocytes. Lymphocytes are part of the immune system, and they help fight infections. Like other blood cells, lymphocytes are made in our bone marrow (the spongy tissue in the center of most bones).

However, in people with ALL, the bone marrow makes too many immature lymphocytes called lymphoblasts (also known as **blasts**). Lymphoblasts can't fight infections, and because they crowd out other blood cells, the bone marrow isn't able to work right.

NCCN experts recommend that people with ALL go to a specialized cancer center for their entire care journey. This includes for evaluation, diagnosis, and treatment.

What's in this guide?

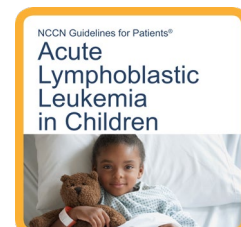
This guide describes everything you need to know about ALL, including:

- Tests used to diagnose ALL and figure out the best treatment for you
- Treatments that are available for the different subtypes of ALL
- The different phases of treatment (there are usually 3: induction therapy, consolidation therapy, and maintenance therapy)
- Supportive care to improve your quality of life

Treating ALL by age: Adult ALL

The information in this guide applies to adults, adolescents, and young adults (sometimes called AYAs) who are being treated for ALL at an adult cancer center. An adolescent and young adult is someone who is 15 to 39 years of age at the time of their cancer diagnosis. Adolescents and young adults are a unique group, and they can be treated in pediatric or adult cancer centers depending on the type of cancer they have.

For children and young adults with ALL who are getting treatment at a pediatric (children's) cancer center, see the *NCCN Guidelines for Patients: Acute Lymphoblastic Leukemia in Children*, available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



What are the types of ALL?

The 2 main subtypes of ALL are B-cell (B-ALL) and T-cell (T-ALL).

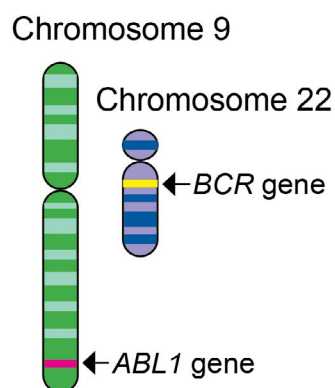
- B-ALL affects B lymphocytes (B cells), and is divided into 2 subtypes:
 - Philadelphia-positive B-ALL
 - Philadelphia-negative B-ALL
- T-ALL affects T lymphocytes (T cells)

Treatments for B-ALL depend on whether leukemia cells have the Philadelphia chromosome, so it's important for your care team to figure out exactly what type of B-ALL you have. The Philadelphia chromosome is an abnormal version of chromosome 22 that contains a fusion gene called the *BCR::ABL1* gene. *BCR::ABL1* is an abnormal gene that's formed when the *ABL1* gene from chromosome 9 joins to the *BCR* gene on chromosome 22. If cancer cells have the Philadelphia chromosome, the type of ALL you have is called **Philadelphia-positive (Ph-positive) B-ALL**. If cancer cells don't have the Philadelphia chromosome, then you have **Philadelphia-negative (Ph-negative) B-ALL**.

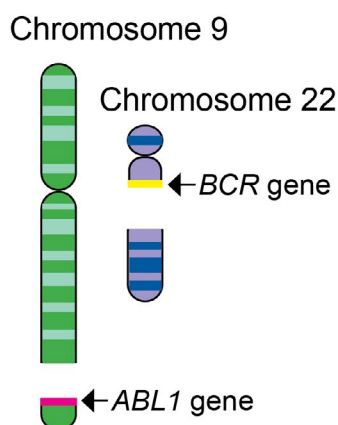
Philadelphia chromosome

The Philadelphia chromosome is created when a piece of chromosome 9 and a piece of chromosome 22 break off and trade places. The result is a fusion gene called *BCR::ABL1* and a shortened chromosome 22 called the Philadelphia (Ph) chromosome.

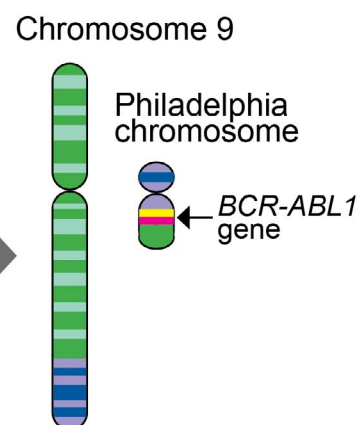
Normal chromosomes



Chromosomes break



Changed chromosomes



How is ALL treated?

The treatment you receive depends on the type of ALL you have, your age, and if you have other health conditions (called comorbidities). All treatment plans for ALL include a type of chemotherapy called intrathecal chemotherapy. This type of chemotherapy is injected into your spinal fluid, and it can reach the brain and spinal cord. It's given during a spinal tap procedure (also called lumbar puncture) to prevent and treat cancer in the nervous system and brain.

Treatment options for ALL usually are a clinical trial (a type of medical research study that looks at new ways to fight cancer) or a combination of systemic therapies (therapies that work throughout the body, such as chemotherapy).

Other treatments include steroids, targeted therapy (treatment that targets specific or unique features of cancer cells), immunotherapy (drugs that use the immune system to fight cancer), CAR T-cell therapy (a type of immunotherapy treatment that's personalized and specialized), and a hematopoietic cell transplant (also known as a stem cell or bone marrow transplant).

People with Ph-positive B-ALL will also receive a type of targeted therapy called a tyrosine kinase inhibitor (TKI). TKIs are given alone or in combination with systemic therapy.

You'll receive treatment in different phases, which are described in *Chapter 4: Phases of treatment*. The 3 main phases are induction therapy, consolidation therapy, and maintenance therapy. Sometimes this can vary, though.

How this guide can help you

Making decisions about cancer care is stressful. There's a lot to learn, and you don't know what the future holds.

Use this guide to get the information and support you need.

Patients, doctors, and other health care professionals trust the NCCN Guidelines for Patients. This guide uses clear, everyday language to explain current cancer care recommendations made by respected experts in the field. Their recommendations are based on the latest research and practices at leading cancer centers.

Your health is unique to you, so your cancer care should be, too. As you read this guide, you'll learn which treatments are likely to provide the best results for you. And you'll be better prepared to talk with your care team.

What is supportive care?

Supportive care is an important part of cancer care. The goal is to improve your quality of life during and after cancer treatment. Supportive care can help manage the side effects of cancer treatment and can also help with other needs, such as paying for care and making treatment decisions. *Chapter 8: Supportive care* describes more about supportive care and what it can do for you.

How can I get the best care?

Advocate for yourself. You have an important role to play in your care. Many people feel more satisfied when they actively take part in planning their cancer care.

The NCCN Guidelines for Patients will help you play a larger role in your care. Discuss the recommendations in this guide with your care team. Ask questions about your options and share your goals and concerns.

Don't know what to ask? You're not alone. That's why we include suggested questions to ask at the end of chapters.

Keep reading to find the best care for you.



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you think!**

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complete an online survey about
the NCCN Guidelines for Patients.**

[NCCN.org/patients/response](https://www.nccn.org/patients/response)

2

Testing to diagnose ALL

- 10 Bone marrow testing
- 11 ALL genetic and molecular testing
- 13 Key points
- 13 Questions to ask

2 Testing to diagnose ALL

Accurate testing is needed to confirm that you have acute lymphoblastic leukemia (ALL). ALL starts in the bone marrow, but it can also involve the lymph nodes; central nervous system (CNS); and organs, such as the liver, spleen, and testicles. This chapter explains the tests and procedures that are used to confirm (diagnose) ALL. Other tests that are used to learn more about the cancer and help plan treatment are described in Chapter 3.

In general, at least 1 out of every 5 cells in your bone marrow must be a lymphoblast (an immature white blood cell) to be diagnosed with acute lymphoblastic leukemia (ALL). ALL

is diagnosed using a bone marrow aspiration and biopsy, immunophenotyping by flow cytometry, and genetic and molecular tests.

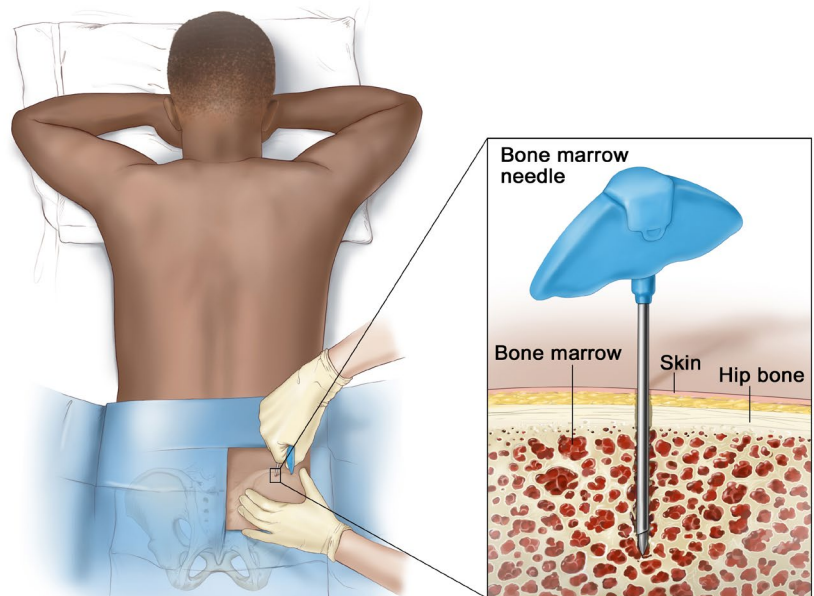
Bone marrow testing

A bone marrow aspiration and biopsy are 2 procedures that are usually done at the same time. A bone marrow aspiration removes liquid bone marrow, and a bone marrow biopsy removes bone marrow tissue. Both samples are usually taken from the back of the hip bone.

The samples are sent to a lab where a pathologist (an expert at using lab tests to diagnose diseases) will examine them. Many different tests will be done on the bone marrow samples, including the ones described next.

Bone marrow aspiration and biopsy

This procedure is done if there might be a problem with blood cells. A bone marrow aspiration and biopsy is used to diagnose ALL, and sometimes it's also used to see how the bone marrow is responding to treatment. It can show if the bone marrow is producing the right amount of certain blood cells.



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Immunophenotyping by flow cytometry

When your bone marrow aspiration and biopsy sample is sent to the lab, an immunophenotyping test will be done on the sample using a technique called **flow cytometry**.

An immunophenotyping test checks for antigens or markers on the surface of cells that can be signs of cancer. If certain antigens are found or aren't found, this can be a sign that the cell is a cancer cell.

Karyotype

A picture of all your chromosomes (called a karyotype or cytogenetics) will be made with a sample of your bone marrow. Normal human cells have 23 pairs of chromosomes, for a total of 46 chromosomes. Leukemia cells can have changes to the normal number or order of chromosomes. A karyotype will show extra, missing, rearranged, or abnormal pieces of chromosomes.

ALL genetic and molecular testing

ALL cells may have changes in genes and chromosomes that can be found with special tests. Genetic and molecular lab tests are used to learn more about the type of ALL you have and to determine the likely path the cancer will take (**prognosis**). Learning more about the cancer will also help your care team plan treatment later.

What are DNA and chromosomes?

Chromosomes contain genetic information and genes (pieces of DNA). Genes give our bodies instructions to make proteins, and proteins make our bodies work the right way. A mutation is when there's a change in the DNA sequence of a cell. Sometimes mutations have no effect on us, other times they're beneficial, and sometimes they're harmful. Sometimes mutations can lead to diseases like cancer, too.

It's important to find out if you have the **Philadelphia chromosome**, which is an abnormal version of chromosome 22 that contains the *BCR::ABL1* fusion gene (described in *Chapter 1: About ALL*). Ph-positive and Ph-negative B-ALL are 2 common subtypes of ALL. There are certain treatments that work best for people with the Philadelphia chromosome, and other treatments that work best for people without the Philadelphia chromosome.

Testing for genetic abnormalities

A sample of your bone marrow or blood will be tested to look for changes or abnormalities (**mutations**) in the leukemia cells. The types of abnormalities found can determine if the cancer is high-risk or low-risk. Some mutations may also determine the type of treatment you need. The lab tests that are done to look for genetic changes are:

2 Testing to diagnose ALL

FISH

Fluorescence in situ hybridization (FISH) testing can find major abnormalities in chromosomes. However, it cannot detect all the possible changes found in genes or chromosomes.

PCR

A polymerase chain reaction (PCR) test creates more copies of DNA so that the DNA can be better examined to find genetic

changes. A reverse transcriptase (RT) PCR test is done for people with ALL to look for the *BCR::ABL1* fusion gene.

Multigene panel testing

Multigene panel testing uses a sample of tissue to look for mutations in genes, such as gene fusions.



ALL genetic changes

Genetic mutations are common in cancer cells. Some genetic changes make cancer respond better to treatment than others. Additionally, certain treatments target specific genetic changes. So it's important to know if you have any genetic changes and, if you do, what they are. Here are some types of genetic changes that can happen:

- **Translocation** – When a chromosome breaks and the broken-off pieces reattach to other chromosomes. For example, the Philadelphia chromosome is a translocation between chromosomes 9 and 22.
- **Deletion** – When a section of DNA is lost or deleted.
- **Rearrangement** – When there is a change in the structure of a chromosome. Pieces of the chromosome could be moved around, missing, or there could be extra pieces.
- **Insertion** – When a new piece of DNA is added.
- **Fusion gene** – When 2 different genes are joined together. This happens when part of the DNA from one chromosome moves to another chromosome. Some people with ALL have the *BCR::ABL1* fusion gene, and treatments can be different for people with and without the gene.
- **Abnormalities in the number of chromosomes** – Sometimes there can be too many or not enough chromosomes.

Key points

- Acute lymphoblastic leukemia (ALL) is diagnosed using a bone marrow aspiration and biopsy, immunophenotyping by flow cytometry, and genetic and molecular tests.
- A bone marrow aspiration and biopsy are 2 procedures that can be done at the same time to remove samples of bone marrow. The samples are usually taken from the back of the hip bone and then sent to a lab. Many different tests are then done on the bone marrow samples.
- An immunophenotyping test (flow cytometry) is done with the bone marrow sample to check for antigens or markers on the surface of cells that can be signs of cancer.
- A picture of all your chromosomes (called a karyotype or cytogenetics) will be made with a sample of your bone marrow. A karyotype shows extra, missing, rearranged, or abnormal pieces of chromosomes.
- The bone marrow sample will also be tested to see if there are genetic abnormalities (mutations) in cells. The tests that will be done are fluorescence in situ hybridization (FISH), polymerase chain reaction (PCR), and multigene panel testing.

Questions to ask

- Who will explain what tests I need and what the results mean?
- What resources are available to help me get to the place I'm getting the test done and to help me pay for the testing?
- What are these procedures like and where are they done near me? Will I have to take the day off from work? Can I receive any pain medications prior to the procedure and, if so, can I drive myself home or do I need a driver?
- How long will it take to get the test results back after I have the bone marrow aspiration and biopsy procedure done?
- Is there a cancer center or hospital nearby that specializes/has expertise in treating ALL? How many people with ALL have you treated in the past year? Am I a candidate for clinical trials and/or a bone marrow transplant and, if so, do I need to go to another center?

3

Testing to plan treatment for ALL

- 15 Medical history
- 15 Physical exam
- 16 Spinal tap with intrathecal chemotherapy
- 17 Central venous access device
- 17 Blood tests
- 19 Imaging scans
- 20 Additional tests that might be needed
- 22 Key points
- 22 Questions to ask

3 Testing to plan treatment for ALL

Your care team needs to learn as much about the cancer as possible to come up with the right treatment plan for you. The tests that are done to find out more about the type of acute lymphoblastic leukemia (ALL) you have and help plan treatment are described in this chapter and listed in **Guide 1**.

Medical history

Your care team needs to have all your health information. They'll ask about the symptoms, health problems, and treatments you've had. A complete report of your health is called a medical history.

When you meet with your care team, be ready to talk about your:

- Illnesses
- Surgeries
- Injuries
- Health conditions
- Symptoms
- Medications and supplements

Family history

Your care team will also ask about the health history of family members who are blood relatives. This information is called a family history. If a family member has had cancer, it's important to know the specific type of cancer, where the cancer started, and if it's in multiple locations. It's also important to find out if they had genetic testing done.

Physical exam

A physical exam is done to look for signs of cancer. During this exam, your doctor or health care professional will check your:

- Temperature, blood pressure, and pulse
- Height and weight

They may also:

- Listen to your lungs, heart, and gut
- Look at your eyes, skin, ears, nose, and mouth
- Feel and apply pressure to parts of your body to see if organs are a normal size or if there's pain when touched
- Do a testicular exam if you were born with male sex organs. A testicular exam is done in certain cases because acute lymphoblastic leukemia (ALL) can sometimes involve the testicles. A scrotal ultrasound may also be done.

Spinal tap with intrathecal chemotherapy

Leukemia can travel to the fluid that surrounds the spinal cord and brain (cerebrospinal fluid). A spinal tap (also called a lumbar puncture)

removes spinal fluid to see if there are cancer cells in those areas.

A spinal tap can also be used to inject chemotherapy into spinal fluid. This is called intrathecal chemotherapy. Everyone with ALL will get intrathecal chemotherapy, even if no

Guide 1

Tests and procedures used to plan treatment for ALL

Medical history and physical exam, including a testicular exam (a scrotal ultrasound may be done, too)

Blood tests

- Complete blood count (CBC), differential, chemistry profile, and liver function tests
- Disseminated intravascular coagulation (DIC) panel to evaluate blood clotting
- Tumor lysis syndrome (TLS) panel to measure levels of lactate dehydrogenase, uric acid, potassium, calcium, and phosphorus
- Hepatitis B, hepatitis C, and HIV tests

Spinal tap (also called lumbar puncture) with intrathecal (IT) chemotherapy

Central venous access device of your choice (allows your care team to give you fluids, blood products, medication, and more)

CT or MRI scan of the head, if you have neurological symptoms (given with contrast)

CT scan of the neck, chest, abdomen, and pelvis or CT/PET scan, if needed (given with contrast)

Strongly consider doing an early evaluation for a hematopoietic cell transplant and looking for a donor

Consider doing heart tests using an echocardiogram or cardiac nuclear medicine scan

Consider cancer risk assessment and counseling, if there may be cancer predisposition syndromes (things in someone's genetic material that make them more likely to develop cancer)

Screening for infections, if needed

Pregnancy testing, fertility counseling, and fertility preservation, if needed

3 Testing to plan treatment for ALL

leukemia cells are found in the spinal fluid, to prevent leukemia cells from spreading there during treatment. Standard chemotherapy (for example, given through an IV or by mouth) usually doesn't penetrate into the spinal fluid or brain. So it's important for chemotherapy to be injected directly into the spinal fluid (even if you don't have symptoms that suggest you need it).

Central venous access device

You may receive a central venous access device, which is a catheter (thin tube) that's inserted into a vein. This allows your care team to give you fluids, blood transfusions, medications like chemotherapy, and more. Different devices include port-a-caths (ports)

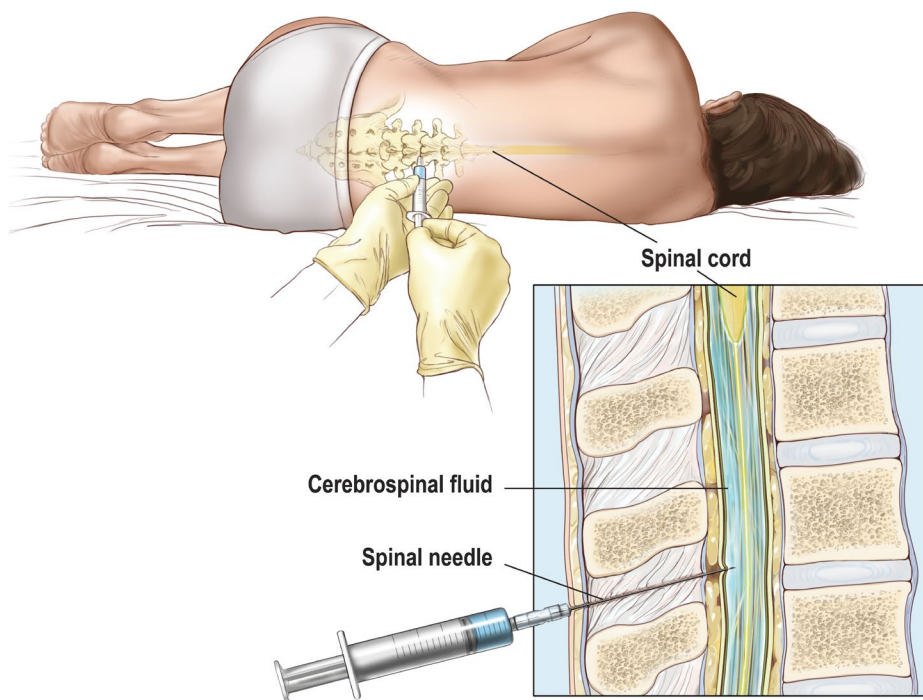
or peripherally inserted central catheter (PICC) lines. There are other types, too. When your care team needs to test your blood, they can also take a sample of blood using the central venous access device.

Blood tests

Blood tests check for signs of disease and can show how well organs are working. They require a sample of your blood, which can be removed by placing a needle into a vein in your arm. Blood tests are also done when you're getting treatment to check blood counts, the health of organs like your liver and kidneys, and to see how the cancer is responding to treatment.

Spinal tap (also called lumbar puncture)

A lumbar puncture is a procedure that removes spinal fluid. The procedure can also be done to give you intrathecal chemotherapy treatment.



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3 Testing to plan treatment for ALL

Some tests that may be done with a sample of your blood are:

Complete blood count with differential

A complete blood count measures the levels of white blood cells, red blood cells, and platelets in your blood. A differential measures the percentage of each type of white blood cell and can also show if there are too many immature white blood cells (called lymphoblasts or blasts). Cancer and other health problems can cause low or high blood cell counts.

Chemistry profile

A chemistry profile or panel measures the levels of different substances in your body, including proteins, enzymes, and sugars. If you have ALL, the chemistry panel results might look different than they usually would.

Liver function tests

Liver function tests look at the health of the liver by measuring substances that are made or processed by the liver. Too much or not enough of certain substances can be signs of an illness, injury, or disease.

Blood clotting tests

Your body stops bleeding by turning blood into a partially solid, gel-like clump called a blood clot. In uncommon cases, leukemia can affect the blood clotting process and cause a blood clotting disorder called disseminated intravascular coagulation (DIC).

To test for this, a DIC panel may be done. This test examines several things that impact blood clotting, including D-dimer level, fibrinogen level, and blood clotting time (prothrombin time and partial thromboplastin time).

Tumor lysis syndrome panel

When treatment kills cancer cells in the body, the substances released by the dead cells sometimes build up faster than the body can get rid of them. This is known as tumor lysis syndrome. It should be treated right away. Changes in lactate dehydrogenase (also called LDH), potassium, phosphorus, calcium, and uric acid levels can be signs of tumor lysis syndrome.

A tumor lysis syndrome panel examines the levels of these substances and other ones, too. Levels are watched closely when you're first diagnosed with ALL, and you might receive medicine and intravenous (IV) fluids to help prevent the levels from getting too high.

Infection and virus testing

Your blood is also tested to see if you've had hepatitis in the past or if you have it now. Some cancer treatments can wake up (or reactivate) hepatitis B virus (HBV) and hepatitis C (HCV) infections, which affect the liver. Your care team may also test you for HIV and COVID-19.

Imaging scans

Imaging scans take pictures of the inside of your body to look for cancer. The images can also provide useful information on where the cancer is and how much of it there is. A radiologist (a doctor who's an expert at reading imaging scans) will review the images and send the results to your care team. These are some of the imaging tests you may have done:

CT scan or MRI scan of the head

You may need a magnetic resonance imaging (MRI) or computed tomography (CT or CAT) scan of your head if you have neurological symptoms. Neurological symptoms, such as headaches, vision changes, tremors, speech changes, and memory problems, can be signs that something is affecting how your nervous system functions.

Both an MRI and CT scan make images of the inside of your body. You'll receive a contrast

agent (also called contrast dye) to make the pictures clearer. Contrast is a substance that's injected into a vein and/or mixed with a liquid to drink, and it leaves your body in urine. Tell your care team if you've had allergic reactions to contrast in the past.

CT scan of the neck, chest, abdomen, and pelvis or PET/CT scan

A CT scan of your neck, chest, abdomen (belly area), and pelvis (hip area) may also be done. In certain cases, a whole-body positron emission tomography (PET) scan is done with a CT scan.

PET scans show whether the cells in your body are functioning normally or abnormally by highlighting cells that may be cancerous. Before you get a PET scan, a substance called a radiotracer is injected into your bloodstream. The radiotracer is detected during the scan and makes cancer cells appear brighter on the

Imaging scans

To look for cancer and figure out where it's spread, you'll have imaging scans done. CT, MRI, and PET scans are imaging options for people with ALL. During the scans you'll lie face-up on a table that moves through a tunnel-like machine. You'll be able to hear and speak with a technician at all times.



3 Testing to plan treatment for ALL

images. Afterward, the radiotracer leaves the body in urine.

Additional tests that might be needed

More tests may be needed to help plan treatment and make sure you get the treatment that's best for you. For ALL, it's common to check for infections and do heart tests. Other tests may be given if you could be pregnant or if you might get a hematopoietic cell transplant.

Heart tests

Heart tests (also called cardiac tests) are used to see how well your heart is working before you start treatment. If you have a heart condition, certain treatments will be avoided.

An echocardiogram (also called an echo) and cardiac nuclear medicine scan are 2 different imaging tests that can be used to check your heart.

- An echocardiogram is an ultrasound of the heart. The echocardiogram uses sound waves to make moving pictures of your heart beating.
- For a cardiac nuclear medicine scan, you'll receive a radiotracer before the scan, and then an imaging machine will take pictures of how the radiotracer moves through your blood and heart.

Checking for infections

When appropriate, screening for infections may also be done (screening is checking for something when you don't have symptoms). To check for infections, your care team may

take another blood sample, do swabs of your nose or throat, or even do more imaging (chest X-ray, CT scans) before treatment starts.

An **opportunistic infection** is an infection that either happens more often or is more severe because the immune system isn't working normally. Treatment for ALL can weaken the body's natural defense against infections. Your care team might monitor you for opportunistic infections and treat you for them if needed.

Importantly, fevers are considered a medical emergency, and you should see a doctor if you develop one.

Cancer risk testing and counseling

Some people have something in their genetic material that makes them more likely to develop cancer. This is known as having a cancer predisposition syndrome. Some predisposition syndromes are hereditary (passed down from parent to child), but others aren't. If your care team thinks you have a cancer predisposition syndrome, they may have you see a genetic counselor.

It's important to find out if you have any cancer predisposition syndromes because they can affect the cancer screenings you get and even the cancer treatment.

Testing if you might get a transplant: HLA typing

Sometimes people with ALL are treated with a hematopoietic cell transplant (also called a bone marrow or stem cell transplant) to kickstart the bone marrow into making healthy blood cells. An allogeneic hemopoietic cell transplant is a type of transplant that replaces

3 Testing to plan treatment for ALL

the unhealthy stem cells in your bone marrow with healthy donated stem cells.

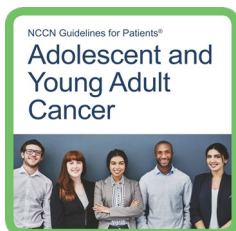
If your care team thinks you may need a hemopoietic cell transplant, they'll do a blood test called human leukocyte antigen (HLA) typing. This test helps find a donor whose stem cells are the best match for you. HLA is different in different people, and a donor has to have an HLA close to your HLA. Finding a donor isn't based on blood type, it's based on HLA type.

Pregnancy test and fertility preservation

Your care team will give you a pregnancy test before treatment begins if there's a chance you could be pregnant. Some cancer treatments can affect your ability to conceive or have children, so your care team may discuss fertility preservation with you before your treatment starts.

The most common fertility preservation options include sperm banking and egg freezing. Sperm banking is when sperm is frozen and stored until after cancer treatment. Similarly, eggs can be removed from ovaries and stored for later use.

More information on fertility preservation can be found in the *NCCN Guidelines for Patients: Adolescent and Young Adult Cancer*, available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



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[NCCN.org/patients/feedback](https://www.nccn.org/patients/feedback)

Key points

- Many different tests are used to help plan treatment for acute lymphoblastic leukemia (ALL). To start learning as much about the cancer as possible, you'll get a physical exam, and your care team will ask about your full medical history.
- A spinal tap (also called a lumbar puncture) is done to see if the cancer has spread to the fluid that surrounds your spinal cord and brain. This procedure removes a sample of the fluid for testing and can also be used to inject chemotherapy into the spinal fluid (intrathecal chemotherapy). Importantly, all treatment plans for ALL include intrathecal chemotherapy, even if no leukemia cells are found in the spinal fluid.
- You'll receive a central venous access device before treatment starts. The device is a thin tube that's inserted into a vein to give you fluids, blood products, medication, and more.
- Different tests will be done on your blood sample. These include a complete blood count with differential, chemistry profile, tumor lysis syndrome panel, infection and virus testing, liver function tests, and blood clotting tests.
- Imaging scans that take pictures of the inside of your body to look for cancer will also be done. You may get a CT or MRI scan of your head if you have neurological symptoms or a CT scan of your neck, chest, belly area, and hip area if needed. A PET/CT scan is given in certain cases, too.

- To help with planning treatment, you may need heart tests and infection screening. If you might receive a hematopoietic cell transplant, an HLA typing test will also be done to help find a donor.

Questions to ask

- Will I need to get any tests or procedures done more than once? How often will my blood be taken for blood tests?
- When will I get the results of all my tests back?
- Can a copy of every test result be printed for me, so I have all my information?
- Can I get a second opinion after I have my test results back?
- Who will talk with me about the next steps and when treatment starts? When can I start treatment?

4

Phases of treatment

- 24 What are the different phases of treatment?
- 24 Induction therapy
- 25 Looking for any leftover cancer cells after treatment
- 25 Consolidation therapy
- 25 Maintenance therapy
- 25 After maintenance therapy: Surveillance and monitoring
- 26 Types of responses to treatment
- 27 Key points
- 27 Questions to ask

You'll be given treatment over different time periods or phases. There are 3 main phases of treatment: induction therapy, consolidation therapy, and maintenance therapy.

What are the different phases of treatment?

The 3 main phases of treatment for acute lymphoblastic leukemia (ALL) are:

- Induction therapy
- Consolidation therapy
- Maintenance therapy

Importantly, not all doctors use the same terms when discussing treatment. The number of treatment phases and the type of chemotherapy you receive depends on the type of ALL you have and how the first phases of treatment work.

Induction therapy

Induction therapy is the first phase of treatment. Treatment is a combination of chemotherapies (called multi-drug chemotherapy). You may spend time in the hospital for part of this treatment.



Main phases of ALL treatment

1. Induction therapy

The first phase of treatment. Induction therapy is done to kill as many cancer cells as possible. The goal of induction therapy is for ALL to go into **remission** (cancer is in remission when the signs and symptoms of cancer decrease or disappear).

2. Consolidation therapy

The second phase of treatment. This is sometimes called intensification therapy. Consolidation therapy is continued treatment given to kill more cancer cells.

3. Maintenance therapy

The last phase of treatment, which is usually the longest. The goal of maintenance therapy is to get rid of all the leukemia in the body.

Looking for any leftover cancer cells after treatment

After induction therapy, your care team will check to see how the cancer responded to treatment by doing a bone marrow test. The treatment plan (treatment regimen) you receive for consolidation therapy depends on if the cancer is in complete remission (meaning there aren't any signs or symptoms of cancer left).

The very small number of cancer cells left in the body during or after cancer treatment are known as **minimal residual disease** (or MRD). Sensitive lab tests can be used to measure the amount of leukemia cells that might be left over after treatment. These tests can detect even 1 leukemia cell out of millions. Although the tests are very sensitive, not all minimal residual disease can be found with tests. This means there may still be leftover leukemia cells even if tests show a negative result.

Minimal residual disease testing might be done after induction therapy and before consolidation therapy. Lab tests will be monitored throughout treatment to see how the cancer is responding to therapy.

Consolidation therapy

After induction therapy, you'll receive more treatment. You'll either get consolidation therapy to treat any leukemia cells that might still be there, or you might get treatment for cancer that hasn't responded to therapy (called refractory disease).

The time spent in the consolidation phase of treatment, and the other phases of treatment too, can vary. The treatments you'll receive are based on factors such as age and how the ALL has responded to treatment.

Maintenance therapy

Maintenance therapy, which is sometimes called continuation therapy, is the final and longest phase of treatment for ALL. Treatment in this phase is less intense than the other phases but it's very important for the long-term cure of ALL.

After maintenance therapy: Surveillance and monitoring

The surveillance and monitoring time period (also called observation) is when your care team checks for any changes in your condition after treatment is completed. Your care team will monitor for cancer and any side effects.

Types of responses to treatment

After treatment, your care team will check to see how the cancer responded to therapy. The cancer can be:

- In remission (cancer went away)
- Relapsed (cancer went away but then came back)
- Refractory (cancer did not get better after treatment)
- Progressing (cancer grew or spread)

Remission

A remission is when the signs and symptoms of a disease, such as cancer, decrease or go away. A remission can be short-term (temporary) or long-term (permanent). A partial remission means the signs and symptoms of cancer have decreased after treatment. A complete remission means all signs and symptoms of cancer have gone away, and the bone marrow is functioning normally again.

When a complete remission happens, all the following are true:

- There are no signs or symptoms of cancer outside the bone marrow (you don't have swollen lymph nodes; a swollen spleen; or cancer involving the skin, gums, testicles, or central nervous system).
- Very few immature blood cells are found. The number of immature cells (called lymphoblasts or blasts) in the bone marrow should be in the normal range (under 5%).
- Blood cell counts have recovered.

Progressive disease

When the amount of ALL increases in your blood or bone marrow during treatment, it's called progressive disease. Disease progression occurs when the cancer spreads to other areas, or when the number of immature blood cells (blasts) in the blood or bone marrow increase by a certain amount.

"While an ALL diagnosis is terrifying, and I never thought I would experience it once, let alone three times, there are options, resources, and excellent treatments out there! I am a firm believer that attitude is everything, and positivity does wonders while undergoing treatment. My advice is always to ask the questions and be your own advocate! "



Key points

- There are 3 main phases of treatment for acute lymphoblastic leukemia (ALL): induction therapy, consolidation therapy, and maintenance therapy.
- Your care team will check on you during treatment and between treatment phases to see how the cancer is responding to therapy.
- Induction therapy is the first phase of treatment. Treatment is a combination of chemotherapies.
- After induction therapy, you'll move onto the consolidation therapy phase of treatment, or you may receive treatment for cancer that hasn't responded to therapy. Consolidation therapy is given to get rid of leukemia cells that may still be there.
- Maintenance therapy is the final, and longest, stage of treatment. It's important for the long-term cure of ALL.
- The surveillance and monitoring time period is when your care team checks on you and looks for any changes in your condition after treatment is finished.

Questions to ask

- How long will I receive induction therapy for?
- How long will it take to do the tests between treatment phases to see how treatment is working?
- Will I be hospitalized for treatment? Can I get any treatments at home?
- Who can help me during each phase of treatment?
- How long is treatment given for overall?

5

Types of treatment

- 29 What is systemic therapy?
- 31 Combination chemotherapy
- 32 Steroids
- 32 Targeted therapy
- 32 Immunotherapy
- 33 Clinical trials
- 35 Hematopoietic cell transplant
- 36 Radiation therapy
- 37 Key points
- 37 Questions to ask

5 Types of treatment

This chapter explains the treatments you might get and how they're given. There are many different types of treatments for acute lymphoblastic leukemia (ALL), and treatment is given in phases. Not everyone receives the same treatment, but all treatment plans include systemic therapy. Other anti-cancer medicines may be part of standard combined chemotherapy treatments, too.

Your care team will look at your test results when making your treatment plan. It's important to have regular talks with your care team about the plan and your goals for treatment.

It's also important to know that treatment for acute lymphoblastic leukemia (ALL) isn't the same for everyone. As the body ages, higher doses of chemotherapy or more intense cancer treatments can become more difficult to tolerate. In addition to age, your overall health, general level of fitness (called performance status), and genetic cancer risk play a role in figuring out what treatments you'll receive.

What is systemic therapy?

Systemic therapy is treatment that targets your entire body. The therapies travel through the bloodstream to treat cancer cells all over the body. Chemotherapy and targeted therapy are types of systemic therapy.

All treatment plans include systemic therapy and/or intrathecal therapy



Systemic therapy is therapy that treats the whole body (such as chemotherapy).



Intrathecal therapy is injected into the cerebrospinal fluid, which surrounds the brain and spine. This allows the therapy to reach these areas, which other treatments can't.

A treatment **regimen** is a plan for treatment that describes the specific dose, schedule, and length of time the therapy or therapies will be given. NCCN experts recommend regimens based on science and safety. When helpful, they assign a level of preference to their recommendations. There are 3 levels of preference:

- **Preferred therapies** have the most evidence they work better and may be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

5 Types of treatment

Later chapters in this guide describe some of the specific treatments that are recommended for the different subtypes of ALL. They will say whether the treatment is a preferred regimen.

Usually, treatments for ALL are a combination of many different drugs, but sometimes a drug is given by itself. The different drugs that can be given alone or as part of combination treatments for ALL are listed in **Guide 2**.

Guide 2

Therapies that may be given during ALL treatment

Chemotherapies	• Mercaptopurine (Purinethol and Purixan) • Vincristine • Methotrexate • Cytarabine • Cyclophosphamide (Frindovyx) • Doxorubicin • Etoposide (Etopophos) • Daunorubicin	• Pegaspargase (Oncaspar) • Idarubicin (Idamycin PFS) • Thioguanine (Tabloid) • Nelarabine (Arranon) • Clofarabine • Fludarabine • Mitoxantrone • Ifosfamide (Ifex)
Steroids	• Prednisone • Dexamethasone	
Targeted therapies	• Inotuzumab ozogamicin (Besponsa) • Venetoclax (Venclexta) • Blinatumomab (Blinicyto) • Rituximab (Rituxan) • Bortezomib (Velcade and Boruzu) • Revumenib (Revuforj)	
TKIs (a type of targeted therapy)	• Asciminib (Scemblix) • Bosutinib (Bosulif) • Dasatinib (Sprycel and Phyrago) • Imatinib (Gleevec and Imkeldi) • Nilotinib (Tasigna and Danziten) • Ponatinib (Iclusig)	
Immunotherapies	• Blinatumomab (considered a targeted therapy and an immunotherapy) • Rituximab (considered a targeted therapy and an immunotherapy)	
CAR T-cell therapies (a type of immunotherapy)	• Tisagenlecleucel (Kymriah) (sometimes called Tisa-cel) • Brexucabtagene autoleucel (Tecartus) (sometimes called Brexu-cel) • Obecabtagene autoleucel (Aucatzyl) (sometimes called Obe-cel)	

Combination chemotherapy

Chemotherapy is a standard treatment for ALL. Many chemotherapies are given together, which is called multi-drug chemotherapy or a multi-agent chemotherapy regimen.

Chemotherapies are often combined because each type works in a different way to kill cancer cells. Chemotherapies can also be combined with other types of treatments (such as targeted therapy). Many chemotherapy drugs are liquids that are injected slowly into a vein (IV infusion). Some are pills or capsules that are swallowed.

Importantly, chemotherapies can cause different side effects. Your care team will check on you throughout treatment to help manage any side effects or uncomfortable feelings.

How often will I get chemotherapy?

Chemotherapy is given in cycles that include treatment days and rest days. This allows your body to recover before you start the next treatment cycle. Cycles vary in length depending on which drugs are used, but a cycle is usually 14, 21, or 28 days. The number of treatment days per cycle and the total number of cycles you'll have varies based on which chemotherapy you receive.

Intrathecal chemotherapy

Treatment plans for ALL include a type of chemotherapy called intrathecal chemotherapy. This type of chemotherapy is injected into your spinal fluid with a spinal tap (also called a lumbar puncture). It's given so the chemotherapy can reach the brain and spinal cord, which other standard chemotherapies aren't able to do. Some treatment plans include intrathecal



Who's on your care team?

Treating ALL takes a team approach. Some members of your care team will be with you throughout your treatment, while others will be there for parts of it. Your team should communicate and work together to bring the best knowledge from each specialty together.

Your care team may include:

- Medical oncologists and hematologists
- Radiation oncologists
- Nurses
- Pharmacists
- Physician assistants
- Patient navigators
- Social workers
- Other specialists



5 Types of treatment

chemotherapy throughout, but others don't include it in maintenance therapy. Because this treatment is given to prevent ALL from spreading to the central nervous system (also called CNS), you may hear it called CNS prophylaxis or prophylactic treatment.

If ALL is found in your central nervous system when you're diagnosed with cancer, you may need more intrathecal chemotherapy or radiation therapy (discussed below) to the brain.

Steroids

Steroids are an important part of ALL treatment plans, and they're usually combined with chemotherapy. Prednisone and dexamethasone are 2 common steroids you may receive.

Targeted therapy

Targeted therapy is a type of treatment that targets specific or unique features of cancer cells. There are many different types of targeted therapy. Some are also considered immunotherapies since they attack cancer cells by enhancing your immune system to fight the cancer or by using antibodies that aren't part of the normal immune system to kill cancer cells.

What's a tyrosine kinase inhibitor?

A tyrosine kinase inhibitor (TKI) is a type of targeted therapy that blocks the signals caused by chemicals, called tyrosine kinases, that make cancer grow and spread. The Philadelphia chromosome is known to cause

a tyrosine kinase called *BCR::ABL*, and there are drugs that can block it. By doing so, they help treat the cancer.

TKI drugs are a little different from one another, but they generally work in a similar way. They may cause different side effects, so you might not be given a certain TKI if you have a particular health condition, such as lung or heart issues.

TKIs are given to treat people with the **Ph-positive B-ALL** subtype of ALL.

Immunotherapy

Immunotherapy is a type of treatment that uses your immune system to find and kill cancer cells. CAR T-cell therapy is a type of immunotherapy that's given to certain people with ALL.

Blinatumomab

Blinatumomab (Blincyto) is a newer drug that's a targeted therapy and immunotherapy. Most people with ALL will be treated with blinatumomab, which is given as an IV infusion. To learn about the side effects of blinatumomab, see *Chapter 8: Supportive care*.

CAR T-cell therapy

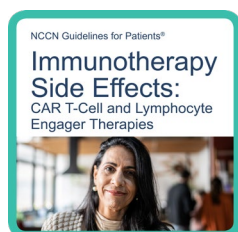
CAR T-cell therapy is a newer treatment that's personalized and specialized, so it may not be available everywhere. CAR T-cell therapy genetically engineers your T cells to fight cancer (T cells are a type of white blood cell that help the body fight diseases).

5 Types of treatment

First, your care team will take some of your blood. Then, the T cells will be separated from the blood, and they'll be sent to a lab where scientists will add something called a chimeric antigen receptor (CAR) to them. Now, the modified T cells (called CAR T-cells) will be able to find and kill cancer cells. Your care team will give you an infusion of the CAR T-cells and they'll start fighting the cancer.

Tisagenlecleucel (Kymriah), brexucabtagene autoleucel (Tecartus), and obecabtagene autoleucel (Aucatzyl) are the currently available CAR T-cell therapy options for people with ALL. But they're only given if the cancer hasn't responded to other treatment, or if it responded to treatment and then came back.

For more information on side effects, see the *NCCN Guidelines for Patients: Immunotherapy Side Effects: CAR T-Cell and Lymphocyte Engager Therapies*, available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Clinical trials

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of fighting cancer need to be studied in people.

If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. FDA.

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- **Phase 1** trials study the safety and side effects of an investigational drug or treatment approach.
- **Phase 2** trials study how well the drug or approach works against a specific type of cancer.
- **Phase 3** trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- **Phase 4** trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It is common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There is no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.



Finding a clinical trial

In the United States

NCCN Cancer Centers
[NCCN.org/cancercenters](https://www.nccn.org/cancercenters)

The National Cancer Institute (NCI)
[cancer.gov/about-cancer/treatment/clinical-trials/search](https://www.cancer.gov/about-cancer/treatment/clinical-trials/search)

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
[cancer.gov/contact](https://www.cancer.gov/contact)

Hematopoietic cell transplant

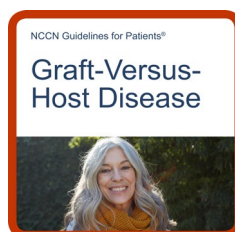
A hematopoietic cell transplant uses chemotherapy or radiation therapy to destroy cancer cells in the bone marrow and then replaces them with new, healthy blood-producing cells (hematopoietic means blood forming). The healthy blood stem cells will grow, form new bone marrow and blood cells, and attack remaining cancer cells.

People with ALL may receive an allogeneic hematopoietic cell transplant. This type of transplant uses healthy blood stem cells from another person (a donor) to regrow your bone marrow. First, you'll receive treatment with chemotherapy and/or radiation therapy. Next, you'll receive an infusion of healthy stem cells from a donor.

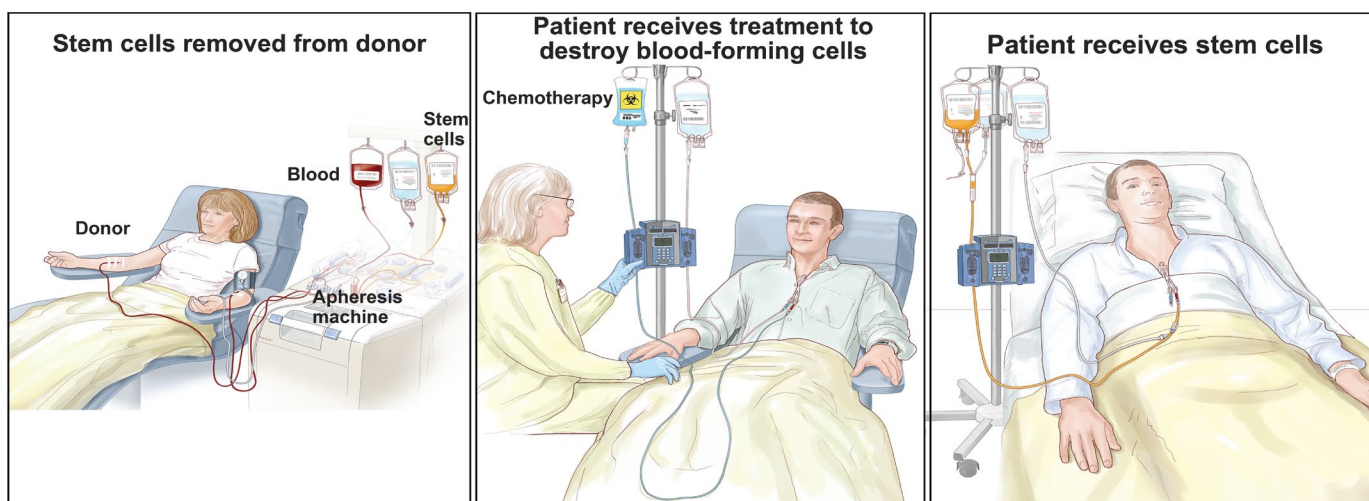
Possible side effects

The side effects of a hematopoietic cell transplant include infections and **graft-versus-host disease (GVHD)**. GVHD is when the donor cells attack your normal, healthy cells. There are treatments for GVHD. Ask your care team about the possible side effects and complications of a hematopoietic cell transplant and how to prevent them.

More information on GVHD can be found in the *NCCN Guidelines for Patients: Graft-Versus-Host Disease*, available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Hematopoietic cell transplant



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Radiation therapy

Radiation therapy uses high-energy radiation beams to kill cancer cells and shrink tumors in certain parts of the body. Radiation therapy may be used for people with ALL who:

- Have cancer in the central nervous system (spinal fluid and/or brain) when they're diagnosed (radiation therapy will be given to the brain area)
- Have cancer in the testicles when they're diagnosed that didn't go away after induction therapy (radiation therapy will be given to the testicles)
- Are in pain or are uncomfortable because of cancer (radiation therapy that may help with pain is called palliative radiation therapy)

When you get radiation therapy, you'll lay on a table, and a large machine will send out radiation beams to a certain part of your body. You won't see, hear, or feel the radiation. It passes through your skin and other tissues to reach the cancer. Radiation therapy usually requires a series of treatment sessions over many days or weeks.

"I quickly learned from my oncologist that he was also a therapist. There's no concern that's too trivial or stupid for them to answer. Their job is so much easier if they know how and what you're feeling."



Key points

- Treatment for acute lymphoblastic leukemia (ALL) isn't the same for everyone, and many different factors will be considered when choosing a treatment for you. But everyone receives systemic therapy, which treats cancer cells all over the body (chemotherapy and targeted therapy are types of systemic therapy).
- NCCN experts recommend systemic therapy treatment plans based on science and safety. They assign the following 3 levels of preference to their recommendations: preferred therapies, other recommended therapies, and therapies used in certain cases.
- ALL is usually treated with a combination of therapies. Chemotherapy is a standard treatment for ALL, and many different chemotherapies are usually combined and given together.
- You'll also receive intrathecal chemotherapy, which is injected into your spinal fluid. It's used so the chemotherapy can reach the brain and spinal cord, which not all other standard chemotherapies are able to reach.
- Many different drugs can be combined with chemotherapy, such as steroids, targeted therapies including tyrosine kinase inhibitors (TKIs), and immunotherapy.
- CAR T-cell therapy is a newer kind of therapy that may be a treatment option for some people. It's a personalized and specialized type of immunotherapy that genetically engineers your T cells (a type of white blood cell) to fight cancer.
- Clinical trials, a hematopoietic cell transplant (also called a bone marrow or stem cell transplant), and radiation therapy are other types of treatment you might receive. Clinical trials are recommended at every stage of ALL treatment but are particularly important when the cancer has come back (relapsed disease) or doesn't respond to initial therapy (refractory disease). Please ask your care team about this option. CAR-T cell therapy, hematopoietic cell transplants, and clinical trials may not be available at your current cancer center, so ask if these are potentially available near you.

Questions to ask

- Which treatment(s) do you recommend and why? What particular chemotherapies would be a good option for me?
- What is your experience treating other people with ALL like me? Have you treated many other people in my situation?
- What clinical trial options are available at your cancer center? Where is the clinical trial treatment given? If one isn't available, can I be seen at another center to discuss this?
- What can I expect from treatment?
- Are there resources to help pay for tests, treatment, or other care I may need?

6

Ph-negative and Ph-positive B-ALL treatments

- 39 What treatments are given for Ph-negative and Ph-positive B-ALL?
- 39 First treatment: Induction therapy
- 40 After induction therapy
- 40 Second treatment: Consolidation therapy
- 41 After consolidation therapy
- 41 Third treatment: Maintenance therapy
- 42 After maintenance therapy: Surveillance and monitoring
- 43 If cancer comes back after it went away or doesn't respond to treatment
- 44 Key points
- 45 Questions to ask

If tests show that you don't have the Philadelphia chromosome or *BCR::ABL1* gene, then the type of acute lymphoblastic leukemia (ALL) you have is called Ph-negative B-ALL (sometimes called Ph-). If tests show that you have the Philadelphia chromosome, then the type of ALL you have is called Ph-positive (sometimes called Ph+) B-ALL. Treatments for these 2 subtypes of B-ALL can be different, and this chapter explains the options.

"Be your own advocate. Talk to someone who has gone through the same thing as you. Ask a lot of questions, even the ones you are afraid to ask. You have to protect yourself and ensure you make the best decisions for you, and get the best care for your particular situation."



What treatments are given for Ph-negative and Ph-positive B-ALL?

Treatments for Ph-negative B-cell acute lymphoblastic leukemia (B-ALL) are usually a combination of systemic therapies (therapies that work throughout the body). Treatments for Ph-positive B-ALL typically include a tyrosine kinase inhibitor (TKI), too. Treatment can be given as part of a standard drug treatment plan (known as a standard-of-care regimen) or as part of a clinical trial when one is available. Different therapies are given during different phases of treatment.

Your care team will tell you which systemic therapy treatments may be best for you. The treatment you get also depends on your age and if you have other health conditions (called comorbidities).

Importantly, everyone with ALL is given intrathecal chemotherapy. This treatment helps prevent cancer in the central nervous system (CNS) and brain.

First treatment: Induction therapy

The first treatment options for **Ph-negative B-ALL** are:

- Treatment in a clinical trial (preferred)
- Systemic therapy

6 Ph-negative and Ph-positive B-ALL treatments

The first treatment options for **Ph-positive B-ALL** are:

- Treatment in a clinical trial (preferred)
- Systemic therapy (such as blinatumomab) plus a tyrosine kinase inhibitor (TKI)

After induction therapy

After you receive induction therapy, your care team will check to see how the cancer responded to treatment. Tests will look to see if the cancer is in complete remission (meaning all the signs and symptoms of cancer have gone away).

If the cancer is in complete remission:

- Your care team will do a test to see if there's any minimal or measurable residual disease (MRD), which is the small number of cancer cells left in the body during or after treatment that can't be seen under a microscope. If no cancer cells are found, your care team will describe the cancer as being MRD negative. If cancer cells are found, it's called MRD-positive cancer. Treatment options can differ depending on if cancer cells are found.
 - If you have **Ph-positive B-ALL** and cancer cells are found, you'll have molecular testing done to look for the Philadelphia chromosome in the leukemia cells. Testing happens before you receive the next phase of treatment and throughout treatment.

If the cancer isn't in complete remission:

- You may have molecular testing done. Then you'll receive more treatment, which is known as treatment for refractory or relapsed cancer (cancer that didn't respond to treatment or cancer that came back after it went away). Options are described later in this chapter on page 43.

After your care team figures out how the cancer responded to treatment, you'll start the next phase of treatment.

Second treatment: Consolidation therapy

If the cancer is in complete remission after induction therapy, consolidation therapy will be given. The consolidation therapy you'll receive is based on if any cancer cells are found.

Treatments given if cancer cells are found (MRD positive)

If you have **Ph-negative B-ALL** and cancer cells are found, treatment options include:

- Treatment with many drugs, including blinatumomab (preferred)
- Blinatumomab

If you have **Ph-positive B-ALL** and cancer cells are found, treatment options include:

- Blinatumomab given with a TKI (preferred)
- Systemic therapy given with many drugs, including a TKI
- A TKI

Treatments given if cancer cells aren't found (MRD negative)

If you have **Ph-negative B-ALL** and cancer cells aren't found, treatment options include:

- Treatment with many drugs, including blinatumomab (preferred)
- Blinatumomab
- Allogeneic hematopoietic cell transplant (also called a bone marrow or stem cell transplant)

If you have **Ph-positive B-ALL** and cancer cells aren't found, treatment options include:

- Blinatumomab given with a TKI
- Systemic therapy given with many drugs, including a TKI
- A TKI
- Allogeneic hematopoietic cell transplant (in some cases)

After consolidation therapy

After consolidation therapy, your care team will do another test to see how the cancer responded to treatment.

If no cancer cells are found (no MRD), then you'll move onto the third phase of treatment, known as maintenance therapy.

If cancer cells are found or if the cancer has spread, you'll get treatment for relapsed or refractory cancer (options are described later in this chapter on page 43). If you have Ph-positive ALL, you'll have more molecular testing done before the treatment, too.

Third treatment: Maintenance therapy

Maintenance therapy is the last phase of treatment. It's given when the cancer is in complete remission, and no cancer cells are found. This is usually the longest phase of therapy, and it's less intense than previous phases.

Maintenance therapy for **Ph-negative B-ALL** is:

- A treatment called POMP (includes the drugs mercaptopurine, vincristine, methotrexate, and prednisone) or an allogeneic hematopoietic cell transplant (transplants are usually given in the consolidation phase of therapy, but they can be given during the maintenance therapy phase, too)

Maintenance therapy for **Ph-positive B-ALL** is:

- A TKI or an allogeneic hematopoietic cell transplant followed by treatment with a TKI

A TKI can be given by itself, or it can be given with vincristine plus prednisone, or with a treatment called POMP (includes the drugs mercaptopurine, vincristine, methotrexate, and prednisone).

After maintenance therapy: Surveillance and monitoring

If the cancer is in complete remission after you receive maintenance therapy or a hematopoietic cell transplant, you'll start the surveillance and monitoring phase of treatment (also called observation). This phase is when

your care team checks on you to make sure the cancer is still in remission. The usual monitoring schedule is explained in **Guide 3**.

If you have Ph-positive cancer, you may receive TKIs and be in the monitoring phase for years or even for the rest of your life.

Guide 3 Surveillance and monitoring for ALL

Usual monitoring schedule:

1 year after treatment	→	Every 1 to 3 months you'll get a: • Physical exam • Blood test (this includes a liver function test until levels are normal and a complete blood count [CBC] with differential)
2 years after treatment	→	Every 3 to 6 months you'll get a: • Physical exam • Blood test (CBC with differential)
3 years after treatment and beyond	→	Every 6 to 12 months, or as needed, you'll get a: • Physical exam • Blood test (CBC with differential)

Additional tests and support:

- Bone marrow aspiration, if your care team thinks it's needed.
- Molecular tests (for people with Ph-positive B-ALL).
- Survivorship recommendations will be given.
- See the *ALL Long-term Follow-up Guidelines* from the Children's Oncology Group (COG) at [survivorshipguidelines.org](https://www.childrensoncologygroup.org/survivorshipguidelines.org).

If cancer comes back after it went away or doesn't respond to treatment

When cancer goes away but then comes back, it's known as a relapse. When cancer isn't getting better after treatment, that means it's refractory to treatment. The treatments for

relapsed or refractory cancer are different for Ph-negative and Ph-positive B-ALL. The Ph-negative B-ALL treatment options are listed in **Guide 4**, and the treatment options for Ph-positive B-ALL are described in **Guide 5**.

A hematopoietic cell transplant can be given in some cases after you receive treatment for relapsed or refractory B-ALL, too.

Guide 4

Treatments for relapsed or refractory Ph-negative B-ALL

- Clinical trial
- Blinatumomab given with or without other therapies (preferred)
- Inotuzumab ozogamicin (preferred)
- CAR T-cell therapy (preferred) with:
 - Brexucabtagene autoleucel
 - Tisagenlecleucel for people under 26 years of age with refractory B-ALL or those who have had 2 or more relapses
 - Obecabtagene autoleucel
- Combination treatment that includes Venetoclax (other recommended)
- Combination treatment that includes alkylator drugs (other recommended)
- Combination treatment that includes Fludarabine (other recommended)
- Combination treatment that includes Cytarabine (other recommended)
- Inotuzumab ozogamicin plus mini-hyperCVD with or without blinatumomab after (other recommended)
- Augmented hyperCVAD (other recommended)
- Clofarabine alone or in combination with other drugs (other recommended)
- MOpAD with Rituximab (for people with CD20-positive ALL) (other recommended)
- Revumenib (for people with a *KMT2A* rearrangement) (other recommended)

Guide 5

Treatments for relapsed or refractory Ph-positive B-ALL

- Clinical trial
-
- TKI given with or without systemic therapy (such as Blinatumomab and Inotuzumab ozogamicin)
-
- CAR T-cell therapy with:
- Brexucabtagene autoleucel for people who just received therapy that included TKIs
 - Tisagenlecleucel for people under 26 years of age with refractory B-ALL, or those who have had 2 or more relapses following treatment that included 2 TKIs
 - Obecabtagene autoleucel for people who just received therapy that included TKIs
-
- The treatment options for Ph-negative B-ALL can also be considered if the cancer hasn't responded to treatment with TKIs (see Guide 4 on page 43)

Key points

- All treatment plans include systemic therapy (therapy that works throughout the body) and intrathecal therapy (injected into the spinal fluid) to prevent cancer in the central nervous system (CNS) and brain.
- Treatments for Ph-negative B-cell acute lymphoblastic leukemia (B-ALL) are usually systemic therapies. Treatments for Ph-positive B-ALL are usually a combination of systemic therapies, including a tyrosine kinase inhibitor (TKI). The systemic therapy you get depends on your age and if you currently have multiple health conditions (called comorbidities).
- Treatment can be given as part of a drug treatment plan (regimen) or as part of a clinical trial when a clinical trial is available. Different therapies are given during different phases of treatment.
- The first treatment (induction therapy) options for Ph-negative B-ALL are treatment in a clinical trial (preferred) and systemic therapy. The options for Ph-positive B-ALL are treatment in a clinical trial (preferred) and systemic therapy plus a TKI.
- After induction therapy, your care team will check to see how the cancer responded to treatment. If the cancer is in complete remission, then the consolidation therapy you'll receive is based on if any cancer cells are found. After your care team sees how the cancer responded to consolidation therapy, maintenance therapy may be given. Maintenance therapy is the last phase of treatment. It's given when the cancer is in complete remission and no cancer cells are found.

- If the cancer is in complete remission after you receive maintenance therapy or a hematopoietic cell transplant, you'll start the monitoring phase of treatment.
- There are also treatment options if the cancer doesn't respond to treatment (refractory) or comes back after it went away (relapsed).

Questions to ask

- Is a clinical trial an option for me? If so, can I get treatment here or do I need to go somewhere else?
- What treatment combination am I going to receive and why? Which TKI do you recommend and why (only applicable to people with Ph-positive B-ALL)?
- Will I get different treatments during each phase of treatment?
- How will you check to see how the cancer responded to treatment? Will I need to have blood tests done?
- What doctor will I see during the monitoring part of treatment? Is it possible to get a detailed calendar or schedule listing all my appointments and chemotherapy treatments to ensure that I take my medications on time and don't miss anything?



Self advocate/ask questions. Your medical team's goal is to save your life. Become informed and also be a partner in your care. Speak up. Now is not the time to be shy. If you don't understand something or want to know if other options are available, ask questions. Don't leave an appointment if you have something you want to say or ask. Don't feel like you should know the answers."

7

T-ALL treatments

- 47 What treatments are given for T-ALL?
- 47 First treatment: Induction therapy
- 47 After induction therapy
- 48 Second treatment: Consolidation therapy
- 48 After consolidation therapy
- 48 Third treatment: Maintenance therapy
- 49 After maintenance therapy: Surveillance and monitoring
- 50 If cancer comes back after it went away or doesn't respond to treatment
- 51 Key points
- 51 Questions to ask

T-cell acute lymphoblastic leukemia (T-ALL) affects T cells rather than B cells. It's less common than B-ALL. This chapter explains the treatment options for T-ALL, which typically include a clinical trial and systemic therapies.

What treatments are given for T-ALL?

Treatments for T-cell acute lymphoblastic leukemia (T-ALL) are usually systemic therapies (therapies that work throughout the body). Treatment can be given as part of a drug treatment plan (regimen) or as part of a clinical trial when one is available. Different therapies are given during different phases of treatment.

The systemic therapy you get depends on your age and if you currently have multiple health conditions (called comorbidities). There are several systemic therapy combinations used to treat ALL, and your care team will tell you which treatment may be best for you.

Importantly, everyone with ALL is given intrathecal chemotherapy. This treatment prevents cancer in the central nervous system (CNS) and brain.

First treatment: Induction therapy

The first treatment options for T-ALL are:

- Treatment in a clinical trial (preferred)
- Systemic therapy

After induction therapy

After you receive induction therapy, your care team will check to see how the cancer responded to treatment. Tests will look to see if the cancer is in complete remission (meaning all the signs and symptoms of cancer have gone away).

If the cancer is in complete remission:

- Your care team will test for any minimal or measurable residual disease (MRD). MRD is the small number of cancer cells left in the body during or after treatment. If no cancer cells are found, your care team will describe the cancer as being MRD negative. If cancer cells are found, it's called MRD-positive cancer. Treatment options can differ depending on if cancer cells are found.

If the cancer isn't in complete remission:

- You'll have genetic testing done and then you'll receive more treatment, which is known as treatment for refractory or relapsed cancer (cancer that didn't respond to treatment or cancer that came back after it went away). Options are described later in this chapter on page 50.

After your care team figures out if the cancer responded to treatment, you'll start the next phase of treatment.

Second treatment: Consolidation therapy

Consolidation therapy is given when the cancer is in complete remission after induction therapy. For T-ALL, this means you'll keep receiving systemic therapy with multiple drugs. Some people may be able to have an allogeneic hematopoietic cell transplant (may be called a bone marrow or stem cell transplant) as their consolidation therapy. Transplants are given as recommended by your care team if the cancer is high risk and may not respond completely to chemotherapy alone in the future.

After consolidation therapy

After consolidation therapy, your care team will do another test to see how the cancer responded to treatment. If no cancer cells are found, then you'll move onto the third phase of treatment, known as maintenance therapy.

If cancer cells are found or if the cancer has spread, you'll get treatment for relapsed or refractory cancer (options are described later in this chapter on page 50).

Third treatment: Maintenance therapy

If the cancer is in complete remission and no cancer cells (MRD) are found, you'll receive the third phase of treatment, maintenance therapy.

Maintenance therapy options are:

- An allogeneic hematopoietic cell transplant
- A treatment called POMP, which includes the drugs mercaptopurine, vincristine, methotrexate, and prednisone

"I learned that getting a second opinion had nothing to do with trusting my doctor. It was the process of getting all the information I needed to make the best decision for my treatment."



After maintenance therapy: Surveillance and monitoring

If the cancer is in complete remission after you've had maintenance therapy or a hematopoietic cell transplant, you'll start the monitoring phase of treatment. Surveillance and monitoring is when your care team checks on you to make sure the cancer is still in remission. The usual monitoring schedule is explained in **Guide 6**.

“

When you're up at 1am, don't Dr. Google. Go to reputable sites that have patient/caregiver information/resources. Recent medically reviewed resources are more valuable than random Google results."

Guide 6 Surveillance and monitoring for T-ALL

Usual monitoring schedule:

1 year after treatment	→	Every 1 to 3 months you'll get a: • Physical exam • Blood test (this includes a liver function test until levels are normal and a complete blood count [CBC] with differential)
2 years after treatment	→	Every 3 to 6 months you'll get a: • Physical exam • Blood test (CBC with differential)
3 years after treatment and beyond	→	Every 6 to 12 months, or as needed, you'll get a: • Physical exam • Blood test (CBC with differential)

Additional tests and support:

- Bone marrow aspiration procedure, if your care team thinks it's needed.
- Survivorship recommendations will be given.
- See the *ALL Long-term Follow-up Guidelines* from the Children's Oncology Group (COG) at survivorshipguidelines.org.

If cancer comes back after it went away or doesn't respond to treatment

When cancer relapses, that means it went away but then came back. When cancer is refractory, that means it isn't getting better after treatment. Treatment options for relapsed or refractory T-ALL include:

- A clinical trial (preferred)
- Systemic therapy

A combination of systemic therapies will likely be given and are explained in **Guide 7**. You may be referred to a specialized center to explore additional treatment options if appropriate.



I remember not being able to physically climb stairs, but today, after treatment, things are getting better and easier to navigate. I can't always see what is around the corner, down below or what will be there when I reach the top, but I still take the steps to get there. One step at a time, one day at a time. I feel like every day since my diagnosis six years ago, I still consciously, every day, make the decision to take 'the steps to get there.'"

Guide 7

Systemic therapy options for relapsed and refractory T-ALL

Other recommended

- Bortezomib with systemic therapy
- High-Dose Cytarabine with systemic therapy
- Mitoxantrone, Etoposide, and Cytarabine
- Nelarabine with or without Etoposide and Cyclophosphamide
- Revumenib (for people with a *KMT2A* genetic rearrangement)
- Venetoclax with systemic therapy
- Systemic therapies used for relapsed/refractory Ph-negative B-ALL may be given, too (see Guide 4 on page 43)

Key points

- All treatment plans include systemic therapy (therapy that works throughout the body) and intrathecal therapy (injected into the spinal fluid) to prevent cancer in the central nervous system (CNS) and brain.
- Treatments for T-cell acute lymphoblastic leukemia (T-ALL) are usually a combination of systemic therapies. The systemic therapy you get depends on your age and if you have other health conditions (called comorbidities).
- Treatment can be given as part of a drug treatment plan (regimen) or as part of a clinical trial when one is available. Different therapies will be given during different phases of treatment.
- If the cancer is in complete remission after induction therapy, consolidation therapy will be given. For T-ALL, this means you'll keep receiving systemic therapy with multiple drugs. Some people may be able to have an allogeneic hematopoietic cell transplant (sometimes called a bone marrow or stem cell transplant).
- If the cancer is in complete remission and no cancer cells are found, you'll receive the third phase of treatment, maintenance therapy. Maintenance therapy options are an allogeneic hematopoietic cell transplant or a treatment called POMP.
- If the cancer is in complete remission after you receive maintenance therapy or a hematopoietic cell transplant, you'll start the monitoring phase of treatment.

- There are also treatment options if the cancer comes back after it went away (relapse), doesn't respond to treatment (refractory), or remains MRD positive (cancer cells are found).

Questions to ask

- What treatment do you recommend based on my age and overall health?
- Is a clinical trial an option for me? If so, can I get treatment here or do I need to go somewhere else? Does another cancer center have a clinical trial I could enroll in?
- Is a hematopoietic cell transplant an option for me?
- Will I need to stay in the hospital during any parts of treatment?
- How will you check to see how the cancer responded to treatment? Will I need to have blood tests done often?

8

Supportive care

- 53 What is supportive care?
- 54 Treating common side effects
- 55 Treating the side effects of certain ALL treatments
- 58 Key points
- 58 Questions to ask

8 Supportive care

Supportive care is given to improve your quality of life. It helps manage the symptoms of acute lymphoblastic leukemia (ALL) and the side effects of treatment. Supportive care can also help with other needs, such as paying for care, getting transportation to care, and more. This chapter talks about the possible side effects of ALL treatments and how they're treated.

What is supportive care?

Supportive care is an important part of cancer care. The goal is to improve your quality of life during and after cancer treatment. Supportive care is for everyone with cancer and their families, not just for those at the end of life.

Supportive care includes a wide range of services. It prevents or manages the symptoms of cancer and the side effects of cancer treatment, like pain and cancer-related fatigue. It also addresses the mental, social, emotional, and spiritual concerns faced by people with cancer.

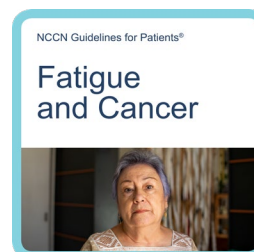
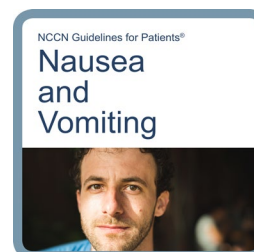
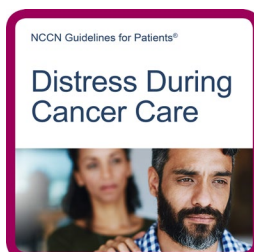
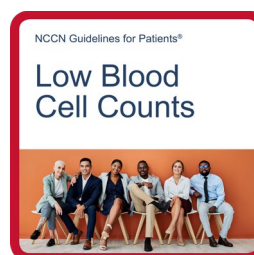
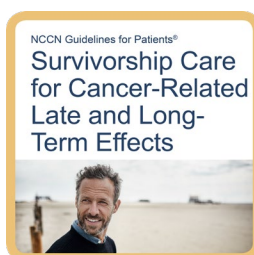
Supportive care provides help with additional needs, such as:

- Making treatment decisions
- Coordinating your care
- Paying for care
- Planning for advance care and end of life

Some health care professionals refer to palliative care as supportive care. They think these types of care are alike or overlap. Others think there are important differences, but both help people with cancer to live as well as they can. Your cancer center may have palliative care specialists, who can give you additional support and help coordinate your supportive care needs.

Supportive care resources

More information on supportive care is available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Read more about the types of support you may receive in the *NCCN Guidelines for Patients: Palliative Care*, available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app. These guidelines also explain more about hospice care, which is care that provides comfort and support at the end of life.

Fevers are considered a medical emergency for people with cancer. It's important to get treatment for a fever if you develop one.

Treating common side effects

All cancer treatments can cause unwanted health issues or side effects. Side effects depend on many factors, such as the particular treatments you get. Some side effects can be harmful to health while others are just unpleasant. Your care team can help you manage side effects, and it's important to tell them if you have any new or worsening symptoms.

These are some common side effects that you could have:

Infections

Cancer treatment can weaken the body's natural defense against infections. Infections happen more often in people with a weakened immune system, and they can be more severe, too. It's important for infections to be treated early before they get worse. Antibiotic medicines may be given to prevent infections.

Your care team will monitor your condition and recommend any vaccines if they're needed. If you have a fever during therapy, you'll need to be treated right away. Contact your care team immediately to start receiving antibiotics and/or to be hospitalized as soon as possible to prevent life-threatening infections and organ damage (sepsis).

Tumor lysis syndrome

Tumor lysis syndrome is a serious side effect of cancer treatment. It happens when the cancer cells being killed by treatment fill the bloodstream with different substances, causing an imbalance in these substances. Symptoms of tumor lysis syndrome include nausea, vomiting, irregular heartbeat, muscle cramps, tiredness, and changes in urination.

There are different treatments for tumor lysis syndrome, such as hydration with IV fluids and drug treatments.

Nutrition and nausea or vomiting

Sometimes side effects from surgery, cancer, or cancer treatment can cause you to lose a significant amount of weight. Speak to your care team if you have trouble eating or if you're losing a lot of weight. They can help to make sure you're getting the nutrition you need. You may also be referred to a nutritionist for help with dietary choices and supplements.

Medicines called antiemetics can also be given to help with nausea or vomiting. Let your care team know if you're having these side effects, and they can help you.

Treating the side effects of certain ALL treatments

This section explains the most common side effects associated with treatments for ALL.

Managing the side effects of steroids

Sometimes steroids, which are part of the chemotherapy treatments (regimens) for ALL, can cause diabetes (high blood sugar levels) even in patients with no history of diabetes. They may also cause mood changes or psychosis (when your mind has trouble connecting with reality) and high blood pressure. Steroids can also cause long-term side effects, which may include injuries to the bones. Your care team will monitor you for these side effects and treat them when needed.

Managing the side effects of treatment with inotuzumab ozogamicin

Treatment with inotuzumab ozogamicin might stop your bone marrow from making enough blood cells or platelets (this is called myelosuppression). Treatment can be given to help. Your care team will also monitor levels of your liver enzymes, because an uncommon liver injury called sinusoidal obstruction syndrome (sometimes called SOS, or veno-occlusive disease) can also occur. The drug ursodiol can be given to prevent SOS in people at risk for it, and the medicine defibrotide can be given to treat this side effect if it happens.

Additionally, certain people who get treatment with inotuzumab ozogamicin also need treatment to reduce the number of immature white blood cells (called lymphoblasts or blasts) in the body (this is sometimes called cytoreductive therapy).

“Leukemia impacts every part of us—physical, emotional, mental, and spiritual. It takes support in all areas to get through the journey. Don’t be afraid to ask for help.”



Managing the side effects of treatment with blinatumomab and CAR T-cell therapy

Treatment with blinatumomab and the CAR T-cell therapies tisagenlecleucel, brexucabtagene autoleucel, and obecabtagene autoleucel can cause conditions called **cytokine release syndrome (CRS)** and **immune effector cell-associated neurotoxicity syndrome (ICANS)**.

CRS is a serious side effect where your immune system releases a lot of cytokines (a protein) into your bloodstream, which can cause inflammation and life-threatening conditions. Your care team will watch closely for it. Signs and symptoms of CRS include fever, chills, muscle aches, nausea, vomiting, headache, fast heartbeat, and trouble breathing. You may be treated with a drug called tocilizumab (Actemra) if you have CRS and other treatments could be given, too. Your care team will check to make sure you don't have an infection, because a serious infection can sometimes look like CRS.

Neurological side effects (ICANS) can also happen if you're treated with blinatumomab or CAR T-cell therapies. These side effects can include confusion, drowsiness, trouble with balance and movement, difficulty finding words, tremors, fainting, and seizures. If you've had CAR T-cell therapy, anti-seizure medicine might be given for a month afterward, and your care team may also give you medicine to help with low blood cell counts.

Treatment with blinatumomab also increases the risk of seizures in people with Down syndrome. Anti-seizure medicine can be given before treatment is started. Additionally, people treated with blinatumomab may need more

treatment to reduce the number of immature white blood cells (called lymphoblasts or blasts) in the body (this is sometimes called cytoreductive therapy).

Pegaspargase and calaspargase chemotherapy side effects

Pegaspargase and calaspargase are a type of chemotherapy called asparaginase chemotherapy. They should only be given in specialized centers and hospitals. The common side effects of these chemotherapies are described below.

Hypersensitivity, allergies, and anaphylaxis

Hypersensitivity is when the immune system overreacts to a drug or substance. This can result in hives, rashes, and trouble breathing. An allergy is when your body reacts to a substance that's normally harmless. An allergic response may cause symptoms such as itching, sneezing, inflammation (swelling), and difficulty breathing or swallowing. Anaphylaxis, or anaphylactic shock, is a severe and possibly life-threatening allergic reaction.

Talk with your care team if you have any allergic reactions. They will monitor you for reactions and treat them when needed.

High blood sugar

Another possible side effect is high blood sugar (also known as hyperglycemia). If you have high blood sugar, insulin will be given to lower these levels.

Organ issues

Your liver and pancreas can be affected by cancer treatment, so your care team will regularly check your organs and run tests to make sure treatment isn't harming them. The treatment you're getting may be stopped

8 Supportive care

permanently or paused until the affected organs are okay again.

Abnormal levels of fats and a blood-clotting protein

Treatment can cause levels of triglycerides (fats) to be too high and levels of fibrinogen (a protein that helps your blood clot) to be too low. These conditions will be treated when needed, and treatment may be paused until they get better.

Hemorrhage

A hemorrhage is bleeding that happens inside or outside the body when a blood vessel is damaged. A hemorrhage can also occur in the head (this is called an intracranial hemorrhage or a brain bleed). If a hemorrhage happens, cancer treatment will be paused, and the hemorrhage will be treated.

Blood clots

Sometimes blood clots can break loose and travel to other parts of the body, blocking a blood vessel (called a thromboembolism). This can cause breathing problems, pain, swelling, or other problems. If you have a thromboembolism, cancer treatment will be paused, and the blood clot will be treated.

Brain blood clot, blood flow issues, or stroke

A brain blood clot (called cerebral thrombosis), blood flow issues (called ischemia), and a stroke are other possible side effects of asparaginase chemotherapy. Treatment will be stopped if needed or paused and then continued when the side effects go away.

"Write everything down/ keep a binder or folder of information. If your loved one wakes up violently ill at midnight, it's after hours, and you need to know who to call. Be prepared before that happens, so you don't scramble in an emergency. Have all phone numbers in one place (even if it's on your phone)."



Key points

- Supportive care is health care that relieves symptoms caused by cancer or its treatment. It also helps improve your quality of life and provides help with other needs. This can include making treatment decisions, coordinating your care, paying for care, getting help with transfusions, and planning for advance care.
- Supportive care can help manage several different side effects, but your particular side effects will depend on many factors, such as the treatments you get. Some side effects can be harmful while others are just unpleasant.
- Common side effects include infections, nausea, vomiting, and nutrition issues. Another possible side effect is tumor lysis syndrome. Symptoms of this syndrome include nausea, vomiting, irregular heartbeat, muscle cramps, tiredness, and changes in urination.
- The side effects of steroids include diabetes (high blood sugar), mood changes, and long-term bone injuries. Your care team will help manage these side effects.
- Inotuzumab ozogamicin, blinatumomab, and CAR T-cell therapies are associated with certain side effects that your care team will look out for. Importantly, treatment with blinatumomab and CAR T-cell therapy might result in a condition called cytokine release syndrome (CRS), which needs to be treated. Symptoms of CRS may include fever, chills, muscle aches, nausea, vomiting, headache, fast heartbeat, and trouble breathing.

- Pegaspargase and calaspargase are 2 chemotherapies that should only be given in specialized centers and hospitals. Your care team will monitor you for certain side effects if you receive either of these drugs.
- Some side effects are very rare. Ask your care team what to expect, and tell them about any new or worsening symptoms you have.

Questions to ask

- What are the side effects of my treatments? How are these side effects managed?
- What should I do if I notice changes in my health?
- Who should I call on nights or weekends if I have side effects, other issues, or a fever? Should I go the emergency room or call 911 if I can't reach a provider?
- Will my care team be able to communicate with the emergency department or urgent care team?
- Can I switch to a different chemotherapy combination if I have a lot of side effects with the one I'm getting?

9

Other resources

- 60 What else to know
- 60 What else to do
- 60 Where to get help
- 61 Questions to ask

Want to learn more? Here's how you can get additional help.

What else to know

This guide helps you know your options so you can make informed decisions and improve your cancer care. But it's not the only resource that you have.

Ask for as much information and help as you need. Many people are interested in learning more about:

- The type of ALL they have
- How often they'll need to get blood transfusions and blood tests done
- What systemic therapy combination treatments they'll receive and what chemotherapies will be part of the treatments
- If they can get CAR T-cell therapy and how much it costs
- How to connect with resources to help with paying for care

What else to do

Your health care center can help you with next steps. It often has on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself.

Where to get help

Look through the list below and visit the provided websites to learn more about these organizations.

AnCan Foundation

[ancan.org](https://www.ancan.org)

Blood Cancer United

[bloodcancerunited.org](https://www.bloodcancerunited.org)

BMT InfoNet

[bmtinfonet.org](https://www.bmtinfonet.org)

CancerCare

[cancercares.org](https://www.cancercares.org)

Cancer Hope Network

[cancerhopenetwork.org](https://www.cancerhopenetwork.org)

GRACE

[cancergrace.org](https://www.cancergrace.org)

Imerman Angels

[imermanangels.org](https://www.imermanangels.org)

Leukemia Research Foundation

[leukemiarf.org](https://www.leukemiarf.org)

National Bone Marrow Transplant Link (nbmtLINK)

nbmtLINK.org

NMDP

nmdp.org/one-on-one

Stupid Cancer

stupidcancer.org

Triage Cancer

triagecancer.org



**Take our survey and help make the
NCCN Guidelines for Patients
better for everyone!**

NCCN.org/patients/comments

Questions to ask

- Who can I talk to about help with housing, food, and other needs?
- How can I connect with others and build a support system?
- How can I reach my doctors after hours or on weekends?
- Can you help me learn more about clinical trials and which ones are available?
- Can I get a second opinion? How do I do that?



Words to know

acute lymphoblastic leukemia (ALL)

A fast-growing blood cancer that starts in the bone marrow and affects a type of white blood cell called lymphocytes.

adolescent and young adult (AYA)

People who are 15 to 39 years of age at the time of their cancer diagnosis.

allogeneic hematopoietic cell transplant

Cancer treatment that destroys cells in the bone marrow and replaces them with healthy blood-producing (hematopoietic) stem cells from a donor.

***BCR::ABL1* gene**

An abnormal gene that is formed when parts of the *BCR* gene and *ABL1* gene join together and create an abnormal chromosome 22, called the Philadelphia chromosome.

blast

An immature lymphocyte (white blood cell). Also called a lymphoblast.

bone marrow

The soft, spongy material inside bones where most blood cells are made.

bone marrow biopsy and aspiration

The removal of a small amount of solid bone marrow and liquid bone marrow to test for cancer.

CAR T-cell therapy

A personalized and specialized type of immunotherapy. CAR T-cell therapy genetically engineers your T cells to fight cancer (T cells are a type of white blood cell that helps the body fight diseases).

chemotherapy

Treatment that kills cancer cells throughout the body.

chromosomes

Long strands of tightly packed DNA inside cells. Chromosomes contain most of the genetic information in a cell.

clinical trial

A type of research that studies investigational tests or treatments in people.

consolidation therapy

The second phase of treatment for ALL.

diagnosis

Identification of an illness based on tests.

DNA

Long strands of genetic information found inside cells.

gene

Sections of DNA that give cells instructions for certain body functions or traits.

hematologist

A doctor who's an expert in blood diseases.

hematopoietic cell transplant

A treatment that replaces damaged or diseased cells in the bone marrow with healthy blood-forming cells. Also called a stem cell transplant or bone marrow transplant.

immunotherapy

A treatment with drugs that use the immune system to fight cancer.

induction therapy

The first phase of treatment for ALL.

maintenance therapy

Usually the last phase of ALL treatment.

minimal residual disease (MRD)

The small amount of cancer cells that are left after treatment. MRD is found by highly sensitive tests done on blood or bone marrow tissue.

mutation

An abnormal change in a cell's DNA.

oncologist

A doctor who's an expert in diagnosing and treating cancer.

Philadelphia chromosome (Ph)

An abnormal chromosome that's made when parts of chromosomes 9 and 22 switch places with each other. The swapping process creates a changed chromosome 22 that now has the *BCR::ABL1* gene on it and is called the Philadelphia chromosome.

prognosis

The likely course and outcome of a disease.

progression

When cancer spreads, grows, or gets worse.

radiation therapy

Treatment that uses high-energy rays (radiation) to kill cancer cells.

refractory cancer

Cancer that doesn't get better with treatment.

regimen

A treatment plan that includes specific information about drug dose, when medicine is taken, and how long treatment will last.

relapsed cancer

When cancer comes back after it had gotten better or gone away.

remission

When the signs and symptoms of cancer decrease or disappear.

steroid

A drug that can be used to reduce inflammation and kill leukemia cells.

supportive care

Health care for the symptoms of cancer and the side effects of cancer treatment. It also addresses mental, social, spiritual, and emotional concerns.

surveillance and monitoring

When your care team checks on you after treatment to make sure there are no signs or symptoms of cancer. Also called observation.

systemic therapy

Treatment that works throughout the body. Chemotherapy and targeted therapy are types of systemic therapy.

targeted therapy

Treatment with drugs that target a specific or unique feature of cancer cells.

transfusion

A medical procedure that involves transferring blood or blood components into a person's bloodstream.

tyrosine kinase inhibitor (TKI)

A type of targeted therapy used to fight cancer.

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Notes

Index

- BCR::ABL1** 6, 11–12, 39
- bone marrow** 5, 7, 10–11, 20–21, 25–26, 35, 41–42, 48–49, 55, 61
- CAR T-cell therapy** 7, 32–33, 43–44, 56, 60
- central nervous system (CNS)** 10, 26, 32, 36, 39, 47
- central venous access device** 16–17
- chemotherapy** 7, 16–17, 24, 29, 30–32, 35, 39, 47–48, 55–57, 60
- chromosome** 6, 11–12, 32, 39–40
- clinical trial** 7, 33–34, 39–40, 43–44, 47, 50
- comorbidities** 7, 39, 47
- consolidation therapy** 5, 7, 24–25, 40–41, 48
- CT scan** 16, 19–20
- diagnose** 5, 10, 18, 32, 36
- DNA** 11–12
- fertility** 16, 21
- flow cytometry** 10–11
- fluorescence in situ hybridization (FISH)** 12
- gene** 6, 11–12
- hematopoietic cell transplant** 7, 16, 20, 35, 41–43, 48–49
- HLA typing** 20–21
- imaging** 19–20
- immune system** 5, 7, 20, 32, 54, 56
- immunophenotyping** 10–11
- immunotherapy** 7, 30, 32–33
- induction therapy** 5, 7, 24–25, 36, 39–40, 47–48
- infection** 5, 16, 18, 20, 35, 54, 56
- intrathecal chemotherapy** 7, 16–17, 31–32, 39, 47
- karyotype** 11
- lymphoblast** 5, 10, 18, 26, 55–56
- lymphocyte** 5, 6, 33
- maintenance therapy** 5, 7, 24–25, 32, 41–42, 48–49
- minimal residual disease (MRD)** 25, 40–41, 47–48
- MRI scan** 16, 19
- multigene panel testing** 12
- mutation** 11–12
- Philadelphia chromosome** 6, 11–12, 32, 39–40
- Ph-negative B-ALL** 6, 11, 39, 40–41, 43
- Ph-positive B-ALL** 6, 7, 32, 39, 40–44
- polymerase chain reaction (PCR)** 12
- prognosis** 11
- radiation therapy** 32, 35–36
- remission** 40–42, 47–49
- spinal tap** 7, 16, 31
- steroid** 7, 30, 32, 55
- supportive care** 5, 8, 32, 53
- surveillance and monitoring** 25, 42, 49
- systemic therapy** 7, 29, 39, 40–41, 44, 47–48, 50, 60
- targeted therapy** 7, 29–32
- transfusion** 17, 60
- tyrosine kinase inhibitor (TKI)** 7, 32, 39–40



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